

E-MAIL SECURITY

- Nowadays both **formal and informal communication** are done through E-mails.
- The personal and confidential communication needs security since it passes through various unsecured channels.
- Malicious software like **viruses, worms and Trojan Horses** can damage the information in various ways.
- Both **Pretty Good Privacy (PGP)** and **Secure/Multipurpose Internet Mail Extensions (S/MIME)** are standards which provide **security to send and receive E-mail in a secure way**.
- E-mail security represents the **collective measures used to protect the access and content of an E-mail account**.
- It permits an **individual or organization to defend the complete access to one or more E-mail account**.

About PGP

- PGP stands for **Pretty Good Privacy (PGP)** which is invented by Phil Zimmermann.
- PGP was designed to provide all four aspects of security, i.e., **privacy, integrity, authentication, and non-repudiation** in the sending of email.

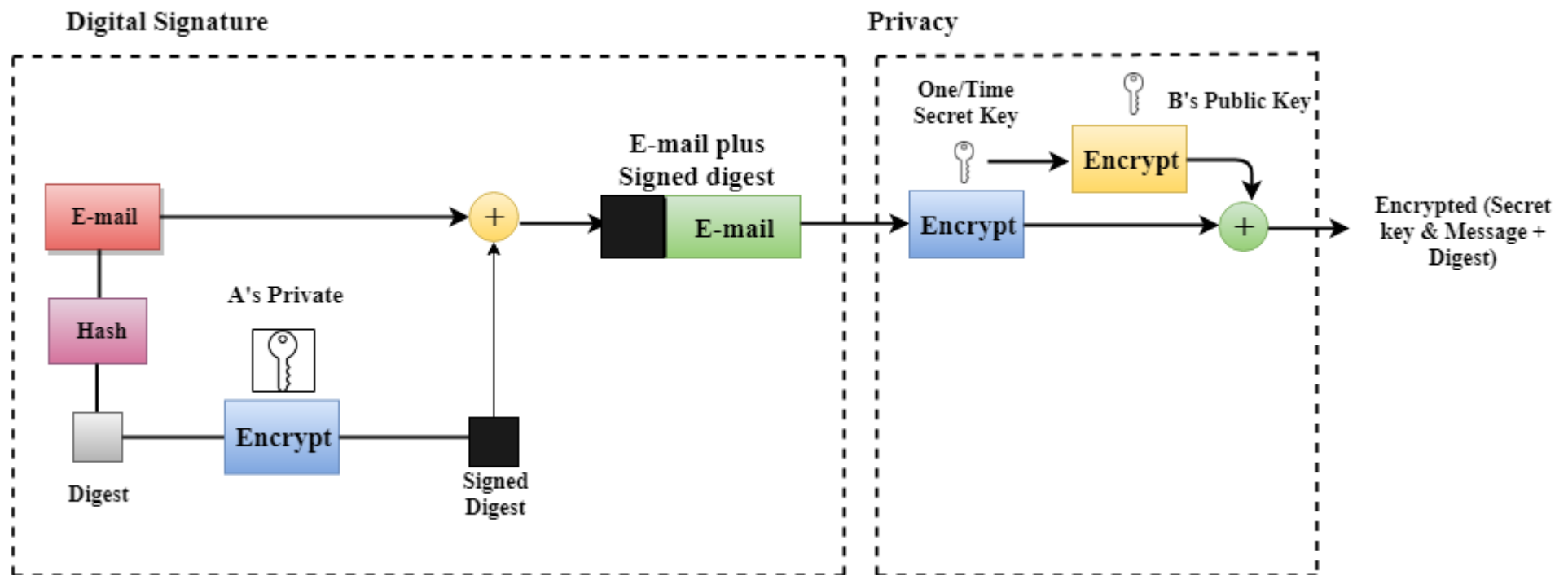
- PGP uses a **digital signature** (a combination of hashing and public key encryption) to provide **integrity, authentication, and non-repudiation**. PGP uses a combination of **secret key encryption and public key encryption** to provide **privacy**. Therefore, the digital signature uses **one hash function, one secret key, and two private-public key pairs**.
- PGP is an open source and freely available software package for email security.
- It provides **compression by using the ZIP algorithm**, and **EMAIL compatibility using the radix-64 encoding scheme**.

Following are the steps taken by PGP to create secure e-mail at the sender site:

1. The e-mail message is **hashed by using a hashing function** to create a digest.
2. The digest is then **encrypted** to form a signed digest by using the **sender's private key**, and then signed digest is added to the original email message.
3. The original message and signed digest are encrypted by using a one-time secret key created by the sender.
4. The secret key is encrypted by using a receiver's public key.

5. Both the encrypted secret key and the encrypted combination of message and digest are sent together.

PGP at the Sender site (A)



Following are the steps taken to show how PGP uses hashing and a combination of three keys to generate the original message:

1. The receiver receives the combination of encrypted secret key and message digest is received.
2. The encrypted secret key is decrypted by using the receiver's private key to get the one-time secret key.
3. The secret key is then used to decrypt the combination of message and digest.
4. The digest is decrypted by using the sender's public key, and the original message is hashed by using a hash function to create a digest.
5. Both the digests are compared if both of them are equal means that all the aspects of security are preserved.

PGP at the Receiver site (B)

